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Validity-Weighted Attenuation of Affective Noise in Open-Ended Teaching Feedback: An Interpretable ABSA Framework

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Abstract
[TODO: FINISH ABSTRACT]

Notes for Practice

- **Users:** instructors and teaching/learning support staff interpreting open feedback.
- **Use:** inspect a review's original text, topics, and adjustment trail; use it to distinguish potentially actionable pedagogical feedback from other student experience signals.
- **Do not use:** automated ranking, promotion, tenure, discipline, or any high-stakes judgment about an instructor.
- **Interpretation:** an attenuated value is a model-based alternative view under a stated relevance taxonomy; it is not a validated estimate of teaching quality.

Keywords
learning analytics; teaching feedback; validity; construct-irrelevant variance; aspect-based sentiment analysis; interpretability

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1. Introduction

1.1 Open-Ended Teaching Feedback as a Learning-Analytics Trace

Open-ended teaching feedback is a widely available but analytically underspecified data source: nothing constrains how the content of accompanying text should shape interpretation of the rating, so any method that attempts to inform stakeholders of its validity must operate under cautious and auditable interpretation. Current methods, however, have been limited to primarily descriptive approaches. Sentiment analysis on Student Evaluations of Teaching (SET) is a growing area for computational insight into student experiences and institutional feedback. SET has historically guided consequential administrative decisions (Haskell, 1997; Simpson & Siguaw, 2000; Irons et al., 2011). Despite this widespread usage, SET validity remains contested: ratings may reflect extraneous influences rather than teaching effectiveness (Langbein, 1994; Marsh & Roche, 1997). Situational factors, irrelevant content, and affective content unrelated to instruction threaten validity (Wongsurawat, 2011; Schiekirka & Raupach, 2015), especially in informal instruments such as RateMyProfessors.com (RMP), which are fully open-ended, anonymized, and largely unmoderated. Interpreting SET results without accounting for contamination risks misinforming stakeholders and distorting faculty assessment (Haskell, 1997; Wongsurawat, 2011; Zumbach & Funke, 2014; Emery et al., 2003; Dowell & Neal, 1983).

Because a numerical rating collapses distinct sources of expression into a single value, the accompanying review text provides a trace through which those sources can be examined. This study treats open-ended teaching feedback as such a learning-analytics trace.

1.2 The Unresolved Methodological Gap

Existing computational SET research provides rich descriptive accounts of sentiment and topics, but offers few mechanisms connecting an explicit relevance taxonomy to a model-estimated adjustment of an accompanying rating. We introduce **validity-weighted attenuation**, an interpretable ABSA method that scales the affect in clauses assigned outside a stated taxonomy while preserving the original review and a review-level audit trail. In this study, *taxonomy-excluded affect* denotes affect expressed in clauses assigned outside the stated taxonomy; the method attenuates its modelled contribution, which is taxonomy-relative and carries no claim that the underlying content is unimportant or invalid. The resulting adjustment makes the sensitivity of a rating summary to the stated taxonomy explicit and auditable, supporting a proposed human-review workflow in a consequential application where important criteria cannot be fully specified (Doshi-Velez & Kim, 2017) and analytical outputs must connect to pedagogical interpretation and action (Gašević et al., 2015). SET provides a consequential context for this proof of concept, and informal platforms such as RMP are a demanding setting for validity-informed feedback analytics. Interpretability is central given concerns about black-box systems in SET (Rybinski & Kopciuszewska, 2021; Zhuang et al., 2025); the framework's audit trail makes its classifications, emotion features, density values, and model-based adjustment available for stakeholder inspection.

Research Questions:

RQ1. What is the topic and affective composition of open-ended teaching feedback, and how do topic-specific emotional signals relate to numerical ratings in the fitted model?

RQ2. How does validity-weighted attenuation change fitted-model outputs and feature associations on held-out instructors, and how stable are these changes under the tested perturbations?

RQ3. How does the mechanism's adjustment ordering align with experts' relative judgments of validity and/or topical relevance on selected review pairs?

To answer our research questions, we develop and apply a five-stage NLP framework to 17,127 RMP student evaluations.

1.3 Contributions

The primary contributions of this research are:

- To our knowledge, the first auditable attenuation mechanism linking an explicit relevance taxonomy to traceable review-level adjustment records in a fitted rating model, motivated by validity and robust-estimation theory (Messick, 1995; Huber, 1964).
- A proof-of-concept characterization of the mechanism on 17,127 RMP reviews, including held-out-instructor analyses and sensitivity checks.
- An initial expert-comparison protocol and density-only reference results that assess agreement with expert rankings on sampled review pairs and establish a concrete baseline for future methods.

The remainder of this paper describes the mechanism and its audit trail (Method), characterizes its behaviour on 17,127 RMP reviews across the three research questions (Results), and interprets the evidence within the stated boundaries: strong coarse-grained alignment with expert judgment, unresolved close-pair performance, and aggregate expert agreement that a density-only ordering reproduces on the same pairs. We regard these results as evidence of mechanistic coherence and as an initial evaluation protocol and reference for future research.

2. Background and Conceptual Framing

2.1 Validity, Construct-Irrelevant Variance, and Teaching Feedback

Messick (1995) identifies two primary threats to validity: construct underrepresentation and construct-irrelevant variance (CIV). His unified theory extends to any assessment in which responses are summarized into scores, a formulation that directly encompasses SET, with these threats particularly pronounced in informal, anonymous instruments such as RMP.

Marsh and Roche (1997) conclude that SET can be multidimensional, reliable, and relatively valid, but condition this on the content and coverage of items: "scores averaged across an ill-defined assortment of items offer no basis for knowing what is being measured" (p. 1187). Validity concerns the evidence supporting score interpretations and uses; the relevance and coverage of elicited content are part of that evidence (Messick, 1995). Unconstrained, open-ended instruments offer no design guarantee that responses adequately represent the intended construct.

Wongsurawat (2011) identifies pedagogically irrelevant comment content as a potential source of measurement error in open-ended SET and argues for evaluating comment reliability. Comments with little to no correlation with average evaluations should be discounted, and administrators would benefit from metrics quantifying comment quality and reliability.

Across these studies, validity threats from construct-irrelevant content, including situational confounders (Schiekirka & Raupach, 2015) and students' emotional and relational states (Li et al., 2025), are documented in both formal and informal evaluation contexts. Recent psychometric work has demonstrated that CIV can be identified and mitigated in constructed-response assessments under controlled conditions (Zhai et al., 2021), but existing approaches either operate under such conditions, which do not extend to large-scale SET, or identify contamination without providing a computational mechanism to attenuate its influence on numerical ratings.

2.2 From Descriptive NLP to Validity-Oriented Interpretation

Prior SET research shows that aggregate sentiment can be associated with numerical ratings (Kechaou et al., 2011; Rajput et al., 2016; Iatrellis et al., 2024; Mihai, 2024), but aggregation obscures topic-specific emotional context. Deep-learning and ABSA (Aspect-Based Sentiment Analysis) techniques enable finer-grained representations of review content (Rybinski & Kopciuszewska, 2021; Shuqin & Raga, 2024); in SET, aspect-based approaches use this granularity to segment feedback and extract aspect-level sentiment (Bhowmik et al., 2023; Nikolic et al., 2020; Jazuli et al., 2025). Ren et al. (2022) extract and aggregate aspect sentiment, while Butt et al. (2025) compare BERT and LLM approaches for SET analysis.

Research on computational estimation of comment relevance shows that filtering by pedagogical value can improve prediction accuracy (Sorour et al., 2015; Louis & Nenkova, 2013; Xiong & Litman, 2011), but does not use relevance to alter affect's contribution to ratings. Neither line of work specifies how pedagogical relevance should modulate affect's contribution to rating interpretation.

2.3 Design Principle: Attenuation Rather Than Deletion

Addressing validity concerns within SET does not require excluding low-utility reviews. Robust estimation motivates down-weighting observations to limit their influence on an estimate (Huber, 1964). We adapt this general principle to topic-specific emotion features, using content density to determine the degree of attenuation.

Pedagogical relevance is operationalized through a generic three-category taxonomy. Clauses assigned to *Miscellaneous* fall outside the included dimensions; because this residual category may include omitted pedagogical concerns, ambiguous wording, or classification error, it provides a basis for inspection rather than a determination of irrelevance. The three categories reflect dimensions that recur across SET instruments and research syntheses (Marsh, 1982), chosen as a generic, widely legible baseline rather than a domain-specific decomposition; different taxonomies would produce different density estimates and adjustments. In formal applications, stakeholders could define or refine the categories in relation to course learning design and its documented pedagogical intent, so that the analytic lens reflects local priorities (Lockyer et al., 2013).

2.4 Research Gap

Despite advances in SET analysis, research offers limited computational support for operationalizing construct relevance to inform transparent, validity-oriented analysis of open-ended teaching feedback and its accompanying ratings. To our knowledge, no prior work links topic-level sentiment to an explicit model-estimated adjustment of accompanying ratings. Such an approach requires prioritizing transparency outputs that are auditable, contestable (Drachsler & Greller, 2016) and connected to pedagogical interpretation and action (Gašević et al., 2015).

3. Method

3.1 Study Context and Data

As a source of data, we use SET reviews from RateMyProfessors.com (RMP), a popular professor and course review website offering anonymous use by any student. As an informal SET system, RMP provides a deliberately noisy setting for examining the framework across heterogeneous and idiosyncratic review content, rather than a proxy for institutional SET. RMP lists a strict "no scraping" policy. Despite attempts to contact site administrators to request permission to source data directly from the site, no permission was granted; we therefore use the open-source dataset provided by He (2020), sourced from RMP. This dataset comprises more than 19,000 individual textual SET reviews and associated numerical ratings (1–5) directed towards hundreds of professors, alongside RMP meta-tags and demographic information measured by the original researchers. For our purposes, the meta-tags and demographic information are not used.

We excluded non-English reviews, reviews containing RMP's automatic censorship string "****" (whose application was prone to error), entries with no review text or text such as "No Comment", and reviews for professors with fewer than eight total reviews. After these exclusions, the final dataset contains 17,127 reviews. Reviews were then cleaned (encoding repair via `ftfy`, lowercase normalization, abbreviation expansion, and punctuation standardization); stemming and lemmatization were withheld to preserve contextual information for downstream models.

3.2 Aspect-Term Categorization

Cleaned text is separated into individual clauses using `spaCy`'s `en_core_web_sm` model, which segments text based on punctuation while attempting to avoid splits on non-statement punctuation. When punctuation is sparse, two additional rules split text on commas and stop-words (but, so, yet), with aggressive stop-word separation used for reviews containing no punctuation.

The next stage assigns each review clause to one of three categories included in this study's pedagogical interpretation—*Instructional Effectiveness*, *Fairness & Grading*, and *Workload & Difficulty*—or to *Miscellaneous*, a residual category for clauses not assigned to those dimensions. Each pedagogical topic is defined using multiple short natural-language descriptions of common student expressions. These descriptions and each review clause are encoded using the pretrained sentence-embedding model `all-mpnet-base-v2`. Each category representation is the elementwise mean of its descriptor embeddings; clause assignment uses cosine similarity against that mean and the clause is assigned as follows:

$$\text{topic}(c) = \begin{cases} \arg\max_{k\in\{\text{eff},\text{fair},\text{work}\}} s_{c,k}, & \text{if } \max_{k\in\{\text{eff},\text{fair},\text{work}\}} s_{c,k} > \tau, \\ \textit{Miscellaneous}, & \text{otherwise}, \end{cases} \tag{1}$$

where $s_{c,k}$ represents the cosine similarity between clause c and pedagogical topic k. We set $\tau = 0.25$ selected on a development set of 386 clauses from 100 manually inspected reviews, drawn before and independently of the evaluation sample, which was matched in size; the two samples share a single review, whose ten clauses appear in both. Threshold selection was thus out-of-sample except for these ten clauses (2.6% of the evaluation sample). By encoding both clause text and category descriptions as semantic vectors, the model captures meaning across the diverse phrasing and informal language common within RMP reviews.

Table 1 summarises the four-category taxonomy with representative descriptor phrases used to construct each topic embedding.

Table 1. Aspect-term categorization taxonomy and representative descriptor phrases.

Category	Representative Descriptors
Instructional Effectiveness	Clear and effective explanations of concepts; lectures are well-organized and easy to follow; helps students learn
Fairness & Grading	Grading feels fair, earned, and based on actual performance; tough but fair grader; adjusts grading to student's current ability
Workload & Difficulty	Heavy workload, lots of effort and time required; tests, exams, or quizzes are hard or demanding; easy class, light workload
Miscellaneous	Clauses below the similarity threshold τ for all pedagogical categories; includes personal remarks and social commentary

3.3 Pedagogical Density and Per-Clause Emotion Extraction

Using the categorized statements, density metrics are calculated for each review:

$$D_{\text{misc}} = \frac{\text{wordcount}_{\text{misc}}}{\text{total wordcount}}, \qquad D_k = \frac{\text{wordcount}_k}{1 + \text{wordcount}_k + \text{wordcount}_{\text{misc}}}, \quad k \in \{\text{eff, fair, work}\}. \tag{2}$$

Here, D_{misc} represents the proportion of content classified as *Miscellaneous* under the stated taxonomy. It operationalizes the share of review text outside this study's included dimensions; the complementary quantity $1 - D_{\text{misc}}$ represents the share assigned to them. This operational measure supports the attenuation rule without establishing the pedagogical relevance of any individual clause. Addition of one guards against division by zero when a pedagogical topic is absent from a review containing no *Miscellaneous* content. The pedagogical density metrics ($D_{\text{effectiveness}}$, D_{fairness}, and D_{workload}) are also used as predictors in regression modelling, but not in the attenuation process.

Per-clause sentiment analysis is performed using `SamLowe/roberta-base-go_emotions` (Lowe, 2024), a fine-tuned RoBERTa model trained on the GoEmotions dataset (Demszky et al., 2020). Each clause c is independently processed and represented by a 27-dimensional emotion vector $\mathbf{e}'_c$, where each component denotes the probability estimated by the model that the clause expresses an emotion label. The model initially produces the 28 GoEmotions labels; *Neutral* is removed because it represents the absence of strong emotional activation rather than an independent affective signal.

Using the previously obtained topic labels, clause-level emotion vectors within each review are aggregated at the topic level by computing the mean emotion vector over all clauses assigned to topic k:

$$\mathbf{E}_k = \frac{1}{|C_k|} \sum_{c \in C_k} \mathbf{e}'_c, \tag{3}$$

where C_k denotes the set of clauses in the review labelled with topic k. If no clauses are assigned to a topic, the corresponding $\mathbf{E}_k$ is imputed as a zero vector of length 27, preserving the fixed input dimensionality while indicating the absence of observed emotion for that topic.

3.4 Predicting Ratings from Topic Specific Emotions

Ratings are modelled as a function of emotions directed towards each topic and density metrics. For a single review, the input feature vector is represented as:

$$\mathbf{x}_{\text{review}} = \begin{bmatrix} \mathbf{E}_{\text{eff}},\ \mathbf{E}_{\text{fair}},\ \mathbf{E}_{\text{work}},\ \mathbf{E}_{\text{misc}}, \\ D_{\text{eff}},\ D_{\text{fair}},\ D_{\text{work}},\ D_{\text{misc}} \end{bmatrix} \in \mathbb{R}^{4\times 27+4} = \mathbb{R}^{112}. \tag{4}$$

Here, $\mathbf{E}_k \in \mathbb{R}^{27}$ is the aggregated emotion vector for topic k, and D_k is the corresponding density metric. For the complete dataset of $N = 17{,}127$ reviews, the feature matrix is $\mathbf{X} \in \mathbb{R}^{N\times 112}$.

To prevent data leakage and assess generalization to unseen professors, we employ an 80–20 professor-level grouped split. The CatBoost regressor is trained on 823 professors and evaluated on 206 held-out professors. Model-performance and coherence analyses use the held-out professors; the expert comparison uses a curated pair sample that spans both groups. This design produces more conservative performance estimates than random splitting while reflecting the challenge of generalizing across professors in larger-scale SET scenarios.

We employ CatBoost regression, an open-source gradient-boosted tree model that captures non-linear interactions between emotional signals and topics in the sparse review representation (Hancock & Khoshgoftaar, 2020; Prokhorenkova et al., 2018; Dorogush et al., 2018). The final configuration was selected through grouped 5-fold cross-validation (GroupKFold, grouped by professor ID) on the training professors: 1000 iterations, learning rate 0.02, depth 7, and L2 leaf regularization 6; all other hyperparameters were kept at their default values. MAE was used as both the loss function and evaluation metric because it measures average absolute deviation from ratings on the 1–5 scale and is resilient to occasional outliers.

3.5 Validity-Weighted Attenuation & Rating Adjustment

Attenuation produces a model-based alternative rating summary by scaling the *Miscellaneous* emotion features according to D_{misc}, the density of review content assigned to that residual category. The down-weighting factor ζ is calculated as:

$$\zeta = 1 - D_{\text{misc}}. \tag{5}$$

A review with no detected *Miscellaneous* content is unchanged ($\zeta = 1$), whereas a review composed entirely of such content receives zero weight for its *Miscellaneous* emotion features ($\zeta = 0$). Intermediate densities produce proportionate attenuation. The factor ζ scales the attenuation of miscellaneous emotion features linearly with their category's share of review text, providing a transparent link between the measured density and the feature transformation.

$$\hat{E}_{\text{misc}}^{\text{weighted}} = E_{\text{misc}} \cdot \zeta. \tag{6}$$

Applying the trained CatBoost model f_{CB}, held fixed across conditions, to the baseline and attenuated feature vectors produces:

$$\begin{aligned} \hat{y}_i^{\text{attenuated}} &= f_{\text{CB}}\left(\hat{\mathbf{x}}_i^{\text{weighted}}\right), \\ \hat{y}_i^{\text{baseline}} &= f_{\text{CB}}(\mathbf{x}_i), \qquad i = 1,\ldots,N. \end{aligned} \tag{7}$$

Adjusted ratings are defined as:

$$\hat{y}_i^{\text{adjusted}} = y_i^{\text{raw}} + \left(\hat{y}_i^{\text{attenuated}} - \hat{y}_i^{\text{baseline}}\right), \quad i = 1,\ldots,N. \tag{8}$$

The prediction difference $\Delta_i = \hat{y}_i^{\text{attenuated}} - \hat{y}_i^{\text{baseline}}$ quantifies the fitted model's response to scaling the *Miscellaneous* emotion features. All other input features remain fixed, while the prediction difference reflects the model's learned relationships and interactions. Adding Δ_i to the raw rating preserves the original prediction residual and yields a model-based alternative summary under the stated attenuation rule.

Because attenuation evaluates the fitted model at transformed feature configurations, its outputs characterize the model's response to the specified transformation rather than ratings that would have been empirically observed. Most sharply, at $D_{\text{misc}} = 1$ all emotion features are zero and every such review receives an identical attenuated prediction, a property of the representation and fitted model rather than an empirically validated affect-neutral target.

3.5.1 Audit Trail and Worked Example

Table 2 presents an end-to-end trace of an individual review with high *Miscellaneous* density (D_{misc}=0.670), where affect in the *Miscellaneous* category dominates, receiving a positive adjustment of +0.427. The trace also makes contestable assignments visible for review: under this generic taxonomy, the classifier assigns interpersonal-conduct language to *Miscellaneous* and subject-knowledge language to *Instructional Effectiveness*.

Table 2. Worked Example, Review #14848.

Clause	Topic	Sim	Top 2 Emotions
dude is huge	*Miscellaneous*	0.030	admiration=0.044, approval=0.024
i mean huge	*Miscellaneous*	-0.010	approval=0.018, annoyance=0.015
is a rude guy	*Miscellaneous*	0.200	anger=0.567, annoyance=0.361
he knows alot about politics	Instr. Effectiveness	0.270	approval=0.293, admiration=0.031

Section 2: Baseline vs. Attenuated Feature Vector

Topic	Baseline Top 4 Emotions	Down-Weighted Top 4 Emotions
Instr. Effectiveness	approval=0.293, admiration=0.031, realization=0.016, optimism=0.004	approval=0.293, admiration=0.031, realization=0.016, optimism=0.004
Miscellaneous	anger=0.192, annoyance=0.129, admiration=0.016, approval=0.016	anger=0.063, annoyance=0.042, admiration=0.005, approval=0.005

Section 3: Validity-Weighted Attenuation

D_{misc}	ζ	Raw	Baseline	Attenuated	Δ	Adjusted
0.670	0.330	3.0	2.698	3.125	+0.427	**3.427**

Three of four clauses are Miscellaneous (D_{misc}=0.670), yielding ζ=0.330. Miscellaneous emotions are down-weighted by approximately 67%. The instructional effectiveness signal remains unchanged. The adjusted rating increases by 0.427.

4. Results

4.1 Corpus and Model Behaviour

4.1.1 ATC Topic Structure

Table 3 summarises topic frequencies after ATC. *Instructional Effectiveness* is the most frequent topic at both clause and review level, followed by *Miscellaneous* and *Workload*, with *Fairness* trailing substantially. The prominence of *Miscellaneous* clauses indicates that a substantial portion of student discourse falls outside this study's included pedagogical dimensions.

The rating distribution is negatively skewed, yet low-rating reviews show distinctive patterns: for ratings 1.0–2.0, *Miscellaneous* clauses are the most frequent topic, exceeding the next pedagogical category by approximately 27% in 1-star reviews.

Figure 1 shows the distribution of D_{misc} across all reviews. The distribution is heavily right-skewed (median 0.11, mean 0.22), with a spike at $D_{\text{misc}} = 1.0$ reflecting reviews composed entirely of content classified as *Miscellaneous* under the stated taxonomy, a subset receiving maximal attenuation.

Table 3. Clause and review-level topic frequencies.

Topic	Clause	Review	Avg./Review
Instructional Eff.	25,845	13,494	**1.92**
Miscellaneous	17,602	10,375	**1.70**
Workload	15,683	9,387	**1.67**
Fairness	4,258	3,567	**1.19**

Together, these patterns show that pedagogical discourse centres on *Instructional Effectiveness*, that *Miscellaneous* content dominates the lowest-rated reviews, and that the right-skewed D_{misc} distribution motivates proportional rather than uniform attenuation.

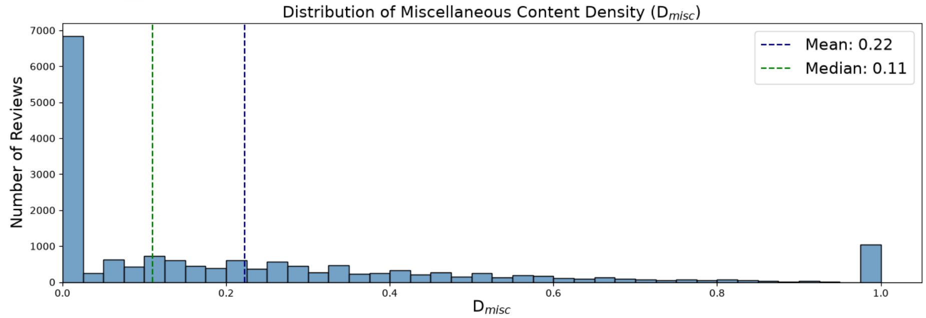

Figure 1. D_{misc} distribution across all reviews.

4.1.2 Sentiment Extraction

RoBERTa-GoEmotions outputs show higher positive-emotion label probabilities in high-rated reviews and higher negative-emotion label probabilities in low-rated reviews, with cross-valence signals at both extremes. One-star reviews also receive positive-emotion label probabilities (Figure 2). These outputs may reflect mixed emotional expression, sarcasm, or classification error.

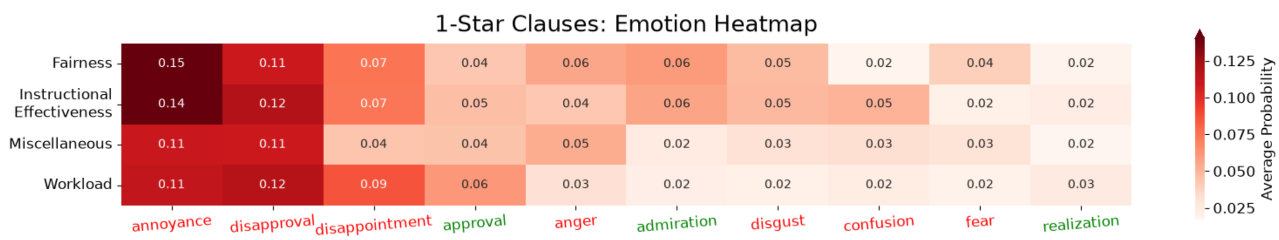

Figure 2. Average emotion probabilities by topic for one-star clauses.

4.1.3 ATC Validation

To characterize inter-model consistency in applying the taxonomy, two independent LLMs (GPT-5 mini and Grok 4.1) categorized 386 clauses, yielding Cohen's (1960) $\kappa = 0.7273$ ("substantial agreement"; Landis & Koch, 1977). This result indicates substantial agreement between the models' category assignments.

A senior faculty expert with more than 15 years of teaching and evaluation experience independently labelled the same 386 clauses, with multiple valid categories permitted per clause. Under a containment criterion (correct if the model's category appeared in the expert's label set), accuracy is 0.77. Restricting to clauses with exactly one expert label ($n = 287$) yields strict accuracy of 0.72; Table 4 reports per-category precision, recall, and F1.

Table 4. Single-label classification performance of ATC against expert annotations.

Category	Precision	Recall	F1	Support
Instr. Eff.	0.73	0.76	0.74	99
Fairness	0.57	0.76	0.65	17
Workload	0.63	0.78	0.69	54
Miscellaneous	0.82	0.66	0.73	117

	Precision	Recall	F1	Support
Macro Avg	0.68	0.74	0.70	287
Weighted Avg	0.74	0.72	0.72	287
Accuracy	0.72 (n = 287)			

These results indicate acceptable alignment for a proof-of-concept study, though accuracy leaves room for improvement, particularly for *Fairness* and *Workload* where support is limited.

4.1.4 Rating Regression Performance

The dataset is split 80–20 by professor to prevent leakage and better reflect the difficulty of generalizing to unseen instructors. Table 5 reports 5-fold GroupKFold CV and held-out performance; the held-out MAE of 0.6347 falls inside the fold range, confirming consistent generalization.

The final model demonstrates low absolute error, strong ordinal consistency, and substantial explained variance on held-out data, suggesting that it captures meaningful relationships between emotional content and student ratings.

SHAP values (Lundberg & Lee, 2017) were computed using exact TreeSHAP on held-out reviews, relative to the model's expected value. For directional inspection, per-review SHAP values were averaged within each of the eight polarity–topic

Table 5. CatBoost 5-fold CV and held-out performance.

	MAE	Spearman ρ	R^2
5-fold CV mean ±SD	0.6300 ±0.0081	0.7284 ±0.0152	0.6255 ±0.0088
Held-out professors	0.6347	0.7352	0.6168

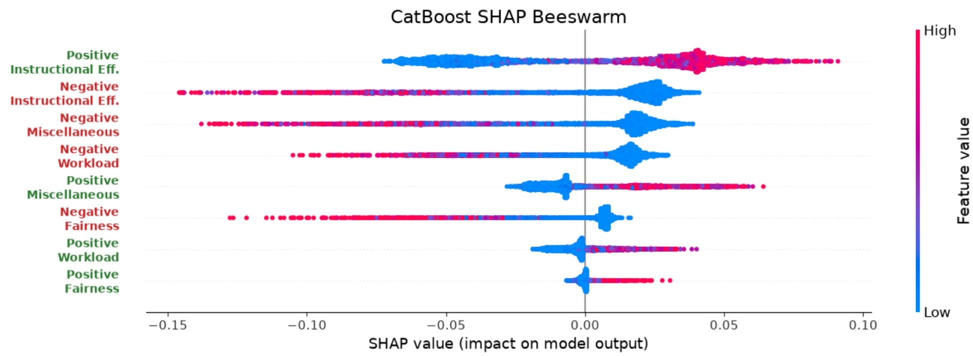

Figure 3. SHAP beeswarm plot for the eight polarity × topic categories on held-out professors. Point colour indicates feature value; horizontal position shows each feature's contribution to the predicted rating.

groups. Beeswarm results show broadly coherent directional patterns: higher positive-emotion feature values generally contribute positively to predicted ratings, while higher negative-emotion feature values generally contribute negatively. At lower feature values, these contribution patterns can reverse (Figure 3).

4.2 What Attenuation Changes

We evaluate whether attenuation produces structured and interpretable changes, examining feature importance shifts, correlation changes, and the relationship between attenuation Δs and modulators (D_{misc}, E_{misc}).

4.2.1 Adjustment Outcomes

Of the 17,127 reviews, 10,375 (60.58%) contained *Miscellaneous* content. Within this subset, the mean $|\Delta|$ was 0.1964 and the mean signed Δ was 0.0440 (Wilcoxon signed-rank $W = 24{,}260{,}091$, $p < .001$); reviews cluster within instructors, so the descriptive pattern rather than the significance level carries the interpretation. The professor-level distribution shape remained largely consistent. Adjustments form a two-sided tail: individual values range from -1.39 to $+2.71$ rating points, with the largest magnitudes concentrated in reviews combining high taxonomy-excluded density with high miscellaneous emotion-label probabilities.

The analyses that follow use the 206 held-out professors: 3,414 reviews for feature-importance analyses and 2,052 reviews containing *Miscellaneous* content for correlation and modulator analyses.

4.2.2 Mechanistic Coherence

Figure 4 and Table 6 summarise three coherence checks on held-out professors. Per-review absolute SHAP-based model contributions were summed within each topic group and averaged across reviews, with changes expressed relative to each group's baseline. Feature importance shifts away from *Miscellaneous* sentiment and toward pedagogically relevant categories (Figure 4a). Pearson and Spearman correlations between baseline emotion features and ratings show the same pattern: *Miscellaneous* associations weaken while pedagogical associations are maintained or slightly strengthened (Figure 4b). These comparisons use the same emotion features before and after attenuation; only the rating changes.

Table 6. Correlation analysis: attenuation Δ versus modulators, computed on held-out professors ($n = 2{,}052$ reviews).

(a) Density modulator D_{misc}

Relationship	Pearson r	Spearman ρ
Δ^+ vs. D_{misc}	0.6872	0.6142
Δ^- vs. D_{misc}	−0.7077	−0.6001

(b) Emotion-signal modulator E_{misc}

Relationship	Pearson r	Spearman ρ
Δ^+ vs. Neg. E_{misc}	0.4008	0.4327
Δ^- vs. Pos. E_{misc}	−0.2470	−0.1125

D_{misc} is the primary driver of attenuation magnitude (Table 6a): adjustment magnitude increases with *Miscellaneous* density, while correlations with miscellaneous emotion features are weaker but directionally consistent. A Mann-Whitney U

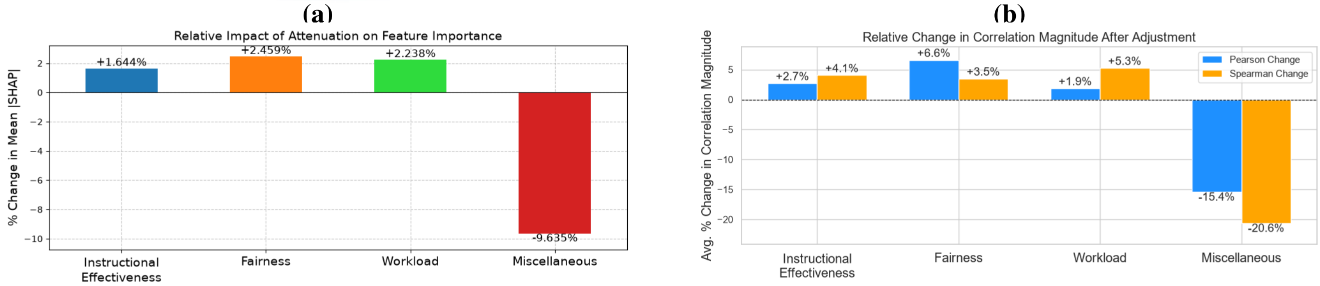

Figure 4. Mechanistic coherence checks after attenuation. (a) Change in SHAP-based feature importance (%), computed on held-out professors ($n = 3{,}414$ reviews). (b) Average change in correlation magnitude (%, Pearson and Spearman) after adjustment, held-out professors ($n = 2{,}052$). Bars show mean $(|\mathrm{corr}_{\mathrm{adjusted}}| - |\mathrm{corr}_{\mathrm{raw}}|)/|\mathrm{corr}_{\mathrm{raw}}|$ across polarity groups.

test (Mann & Whitney, 1947) with rank-biserial effect size (Cureton, 1956) confirms significantly larger absolute adjustments for reviews with higher *Miscellaneous* density ($D_{\mathrm{misc}} > 0.5$, $n = 508$) than for reviews with lower density ($D_{\mathrm{misc}} < 0.2$, $n = 690$) ($U = 313{,}494$, $p < .001$, $|r| = .789$).Reviews cluster within instructors, and some instructors appear in both density groups, so the effect size rather than the significance level carries the interpretation. Together, these checks indicate that attenuation introduces structured and interpretable adjustments consistent with its design rationale.

4.3 Initial Validity and Robustness Evidence

4.3.1 Expert Paired Comparisons

Three senior faculty experts with combined 75 years of experience in teaching, evaluation, and supervision independently labelled the same 80 manually anonymized RMP review pairs. Each pair presented two review texts; instructor and institution names were replaced with tags, and profanity and emoji were replaced where present. Pair order was randomized once and retained across experts. 3 pairs with no model outputs were also included to further blind the experts. Experts selected review 1 when it was "more valid and/or on topic," review 2 when the second review was, and 3 only when completely unsure. The prompt left these terms to the experts' professional judgment and did not provide the study taxonomy. Experts were blind to model outputs and one another's labels. The verbatim task description text is provided in the supplementary material; all three experts received identical prompts. Pairs were constructed by randomly sampling an initial review, informally checking its ATC assignment, and sampling a second review until the required adjustment-contrast bin was met. Candidate rejection counts during pair construction were not recorded. Excluding the three pairs without model outputs, there were 77 evaluable pairs: 43 coarse and 34 fine. Coarse pairs had an absolute-adjustment difference $\big||\Delta_1| - |\Delta_2|\big| \geq 0.5$; fine pairs had a difference below 0.5. At the operating point $s = 1$, these conditions partition the evaluable pairs. The 77 pairs reference 144 distinct instructors; eight instructors appear in more than one pair (at most three), and no pair contains two reviews of the same instructor.

Two matching votes established a majority. A model ranking agreed when the review receiving smaller $|\Delta|$ matched a directional majority; exact model ties selected the first displayed review. This comparison tests whether ordering reviews by the magnitude of the fitted model's response to taxonomy-based attenuation aligns with experts' judgments of validity and/or topical relevance. All retained pairs had three votes and a majority; one "unsure" majority counted as incorrect. Table 7 reports majority agreement, two-sided Wilson (1927) intervals, and Fleiss' (1971) κ over all three response categories. Following (Landis & Koch, 1977), these correspond to substantial ($\kappa = .658$, overall), almost perfect ($\kappa = .819$, coarse), and moderate ($\kappa = .469$, fine) agreement. Individual-expert fine agreement ranged from 58.8% to 70.6%. As a sensitivity check, excluding individual non-directional votes raised fine inter-expert agreement from $\kappa = .469$ to generalized $\kappa = .634$ ("substantial agreement"); this describes agreement among retained directional votes and does not change the primary scoring.

For the density-only baseline, exact density ties selected the first displayed review, matching the attenuation ranking's tie rule. Two density ties occurred; counting these as incorrect instead left agreement unchanged.

Density alone matched attenuation's agreement overall and within both contrast groups. Both methods correctly ranked 61 pairs; each correctly ranked three additional pairs that the other did not. Thus, this sample shows no aggregate agreement advantage for attenuation over density alone, although the methods do not succeed on identical sets of pairs.

A post hoc diagnostic partitioned the 34 fine pairs by expert voting pattern: of 19 unanimous pairs, 16 agreed with the model ranking (84.2%); of 15 split-vote pairs, 5 agreed (33.3%). Ten of thirteen model–majority disagreements occurred on pairs where the experts themselves disagreed. Three disagreements persisted despite expert unanimity, at absolute-adjustment contrasts of 0.314, 0.005, and 0.026 points.

Table 7. Attenuation and density-only rankings compared with three-expert judgments. Attenuation selects the review with smaller $|\Delta|$; density alone selects the review with lower D_{misc}. Both methods use the same 77 evaluable pairs and the original attenuation-defined coarse/fine groups. Agreement requires a matching directional majority; absent or non-directional majorities count as incorrect. Agreement and two-sided 95% Wilson intervals are identical for the two methods within each condition. Expert κ measures agreement among experts over all three response categories (review 1, review 2, "unsure"), using all complete three-rater pairs in each condition (77 overall, 43 coarse, 34 fine).

Condition	Attenuation correct/n	Density-only correct/n	Agreement for each	95% CI for each	Expert κ
Overall	64/77	64/77	83.1%	[73.2%, 89.9%]	.658
Coarse	43/43	43/43	100%	[91.8%, 100%]	.819
Fine	21/34	21/34	61.8%	[45.0%, 76.1%]	.469

4.3.2 Permutation Control

For each of 1,000 permutations (Ernst, 2004), we shuffled the positive D_{misc} values across the full dataset of 17,127 reviews and recomputed the attenuated ratings with the fitted model held fixed. The shuffled density supplied both a model feature and the attenuation input. Expert-pair agreement was then evaluated on the same 77 pairs. Modulator correlations were evaluated separately on the 2,052 held-out reviews with positive D_{misc}, using each review's original density as the reference. The primary expert test recomputed coarse/fine membership after each shuffle; a supplementary test kept the observed $s = 1$ membership fixed. Table 8 reports both specifications.

Table 8. Permutation control ($N = 1{,}000$). Expert rows report agreement; modulator rows report Pearson r. Null entries give the mean and one-sided p using $(k+1)/(N+1)$ (Phipson & Smyth, 2010). Recomputed coarse groups contained a mean of 10.51 pairs (range 3–20), compared with 43 observed; fine groups averaged 66.49, compared with 34 observed. Fixed-bin subgroup analyses are supplementary; overall and modulator tests are shared.

Metric	Observed	Recomputed null (mean, p)	Fixed-bin null (mean, p)
Expert agreement (overall)	83.1%	63.3%, .001	Same test
Expert agreement (coarse)	100%	84.9%, .180	68.9%, .001
Expert agreement (fine)	61.8%	59.9%, .378	56.3%, .321
Δ^+ vs. D_{misc} (r)	.6872	.2327, .001	Same test
Δ^- vs. D_{misc} (r)	−.7077	−.1579, .001	Same test

Overall expert agreement exceeded the null mean by approximately four null standard deviations (83.1% vs. 63.3%, $p = .001$); the null never reached the observed value in 1,000 shuffles. Under shuffled density, the coarse/fine partition was not preserved: the recomputed coarse group averaged 10.5 pairs with changing membership, and 179 shuffles reached 100% agreement ($p = .180$), so the recomputed-bin null evaluates smaller, shifting groups rather than the observed pairs. On the fixed 43 coarse pairs, no shuffle matched the observed agreement ($p = .001$; null maximum .884), indicating that the observed density assignment specifically contributes to the model's ranking of high-contrast pairs. Neither fine specification rejected its null ($p = .378$ recomputed, $p = .321$ fixed), with observed advantages over the null means of 1.9 and 5.5 percentage points; close-pair alignment is therefore not established at this sample size.

No shuffle produced a positive modulator correlation as large, or a negative correlation as negative, as observed ($p = .001$ each). These results support dependence on the original density assignment under the specified perturbation, conditional on the fitted model and classifications.

4.3.3 Sensitivity to Attenuation Exponent

We examine the attenuation family $\zeta = 1 - D_{\text{misc}}^{s}$ around the operating point $s = 1$, which makes feature attenuation directly proportional to miscellaneous content density. Values below and above one provide alternative weighting curves for assessing sensitivity to this design choice. We evaluate $s \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$.

Table 9(a) reports selected delta comparisons: the two adjacent contrasts around $s = 1$ and the full tested span. Adjacent values yield near-identical adjustments (e.g., $s = 0.75$ vs. 1.0: $r = 0.995$, mean $|\Delta_a - \Delta_b| = 0.029$), and even the most distant pair ($s = 0.5$ vs. 1.5) retains $r = 0.964$, although rank agreement is lower ($\rho = 0.689$). The maximum single-review difference for that pair is 1.148, but this is one extreme case; the mean across all 2,052 reviews is 0.083.

Table 9(c) shows that absolute Spearman correlations increase monotonically with s across the tested range, while Pearson correlations remain nearly constant. The positive and negative relationships retain their respective directions throughout. These

Table 9. Robustness across attenuation exponents: (a) selected delta comparisons, with $d = |\Delta_a - \Delta_b|$; (b) expert majority agreement, with the same pairs and bins fixed at $s = 1$ ($n = 77$: 43 coarse, 34 fine; operating point in bold); (c) modulator correlations; (d) professor-bootstrap 95% confidence intervals. Panels (a), (c), and (d) use 2,052 held-out reviews with $D_{\text{misc}} > 0$; panel (b) uses the full expert sample.

(a) Pairwise delta agreement

s_a	s_b	r	ρ	Mean d	Max d
0.75	1.00	0.995	0.920	0.029	0.429
1.00	1.25	0.997	0.917	0.022	0.265
0.50	1.50	0.964	0.689	0.083	1.148

(b) Expert majority agreement

s	Overall	Coarse	Fine
0.50	88.3%	100%	73.5%
0.75	88.3%	100%	73.5%
1.00	**83.1%**	**100%**	**61.8%**
1.25	87.0%	100%	70.6%
1.50	87.0%	100%	70.6%

(c) Modulator correlations

s	$\Delta^+ r$	$\Delta^+ \rho$	$\Delta^- r$	$\Delta^- \rho$
0.50	0.677	0.505	−0.688	−0.502
0.75	0.684	0.576	−0.703	−0.542
1.00	0.687	0.614	−0.708	−0.600
1.25	0.686	0.656	−0.709	−0.646
1.50	0.677	0.680	−0.717	−0.694

(d) Bootstrap professor stability

Metric	Mean	SD	95% CI	
$\Delta^+ D_{\text{misc}}$ (r)	0.687	0.017	0.651	0.720
$\Delta^- D_{\text{misc}}$ (r)	−0.707	0.018	−0.741	−0.671
$\Delta^+ D_{\text{misc}}$ (ρ)	0.613	0.027	0.557	0.661
$\Delta^- D_{\text{misc}}$ (ρ)	−0.601	0.025	−0.649	−0.550

results describe sensitivity of the fitted mechanism to the attenuation exponent.

Table 9(b) holds expert-pair membership and coarse/fine bins fixed at $s = 1$, as in the primary expert analysis. Overall agreement ranged from 83.1% to 88.3% across the tested exponents ($n = 77$). Coarse agreement remained 100%, while fine agreement varied from 61.8% to 73.5% (21–25 of 34 pairs).

Table 9(d) reports 95% confidence intervals from 1,000 bootstrap resamples of held-out professors. Across resamples, the four correlations had SDs of 0.027 or less, indicating stability across professor samples conditional on the fitted pipeline.

5. Discussion

5.1 A Validity-Oriented Analytic Primitive

RQ1 results show that *Instructional Effectiveness* is the primary driver of rating variation in the fitted model, dominating in both frequency and attribution. Affect in the *Miscellaneous* category ranks second only to *Instructional Effectiveness*, exceeding both *Fairness* and *Workload*, motivating an auditable method for examining its modelled contribution. Under the stated taxonomy, such content may introduce construct-irrelevant variance into rating interpretation, consistent with broader concerns about non-instructional influences on teaching evaluations (Marsh & Roche, 1997; Langbein, 1994; Schiekirka & Raupach, 2015; Dowell & Neal, 1983; Wongsurawat, 2011; Li et al., 2025). Rating extremes display different profiles: concentrated positive emotions in 5-star reviews versus diffuse negative emotions across all topics in 1-star reviews, with *Fairness* disproportionately represented in low ratings.

Addressing RQ2, held-out coherence checks show that attenuation reduces model-estimated feature attribution associated with taxonomy-excluded affect (−9.6%) while shifting attribution toward pedagogical categories (+1.6% to +2.5%); correlations with pedagogical categories are maintained or slightly strengthened. Adjustment magnitude tracks *Miscellaneous* density in both directions, with weaker but directionally consistent relationships with miscellaneous emotion features. Within reviews containing *Miscellaneous* content, the mean absolute adjustment is 0.1964 points. The two-sided tail arises at the mechanism's $D_{\text{misc}} = 1$ collapse point: all *Miscellaneous* emotion features are zeroed, the attenuated prediction becomes the fixed baseline $\hat{y} = 3.65$, and Δ equals the difference between the review's baseline prediction and that fixed prediction. The observed extremes are $+2.71$ and -1.39 rating points.

These patterns are stable under the tested perturbations. The permutation control confirms that the modulator correlations depend on the original density assignment: no shuffle produced correlations as extreme as observed ($p = .001$ each), with the observed values ($r = .687, -.708$) more extreme than the corresponding permuted means ($r = .233, -.158$), conditional on the fitted model and the specified perturbation, in which density was shuffled both as a model feature and as the attenuation input. Professor-level bootstrap CIs ([.651, .720] and [−.741, −.671]) confirm stability across held-out professor samples. Across attenuation exponents $s \in \{0.5, \ldots, 1.5\}$, Pearson modulator correlations vary little. Around the operating point, adjacent exponents yield near-identical adjustments: $s = 0.75$ versus 1.0 gives $r = .995$ with mean $|\Delta_a - \Delta_b| = .029$, and $s = 1.0$ versus 1.25 gives $r = .997$ with mean $|\Delta_a - \Delta_b| = .022$; larger separations produce lower rank agreement. $s = 1$ was selected a priori on proportionality grounds. The Mann–Whitney rank-biserial ($|r| = .789$), computed on deliberately separated density groups ($D_{\text{misc}} > 0.5$ vs. $D_{\text{misc}} < 0.2$), indicates that a high-density review receives a larger-magnitude adjustment than a low-density review in approximately 89% of cross-group pairings. Together, these checks support coherence across held-out professor resamples and the tested attenuation exponents, while the permutation result shows that the observed pattern depends on the original density assignment.

Addressing RQ3, adjustment-magnitude ordering agreed with experts' relative judgments of validity and/or topical relevance on 83.1% of 77 sampled pairs (100% coarse, 61.8% fine; 43 of the 64 correct pairs are coarse). Overall agreement exceeded the permutation benchmark ($p = .001$); the null never reached the observed value in 1,000 shuffles. In the supplementary fixed-bin analysis, no shuffle matched the observed agreement on the original 43 coarse pairs ($p = .001$), indicating that the original density assignment contributes to the model's ranking of high-contrast pairs. Fine-condition agreement did not exceed the permutation null under either bin specification ($p = .378$ recomputed, $p = .321$ fixed); close-pair alignment is not established at this sample size. Across the tested exponents, fine-pair agreement ranged from 61.8% to 73.5%.

Density alone matched attenuation's aggregate agreement on these sampled pairs, overall and within both contrast groups. Both methods correctly ranked 61 pairs; each correctly ranked three additional pairs that the other did not. This evaluation provides no observed agreement advantage for the full mechanism over the density-only ordering on these pairs. The comparison establishes an initial reference for subsequent methods on this curated sample.

A post hoc unanimity breakdown showed that fine disagreements were concentrated in split-vote pairs (10 of 13 model–majority disagreements), although three persisted despite expert unanimity. The study cannot determine whether these reflect limits of the mechanism in close cases or differences between the experts' broader judgments of validity and/or topical relevance and this study's taxonomy. The fine-condition results and the density-only match provide an initial reference for evaluating future methods' agreement with expert judgments. The model analyses additionally characterize learned feature–rating relationships and responses to attenuation; whether presenting these outputs helps stakeholders interpret teaching feedback requires separate evaluation.

The framework addresses similar CIV concerns to those documented by Zhai et al. (2021) in controlled assessments, but in the considerably more challenging setting of informal, unmoderated SET reviews. This positions the work as initial method evidence for a validity-oriented direction in teaching-feedback analytics, rather than a ready intervention.

5.2 Implications for Responsible Interpretation of Teaching Feedback

At corpus scale, the framework can generate an auditable review queue that helps reviewers prioritize their inspection across large collections of feedback. Sorting reviews by the absolute difference between raw and adjusted ratings, $|\Delta|$, highlights cases in which taxonomy-excluded affect materially changes the modelled summary. These cases can be prioritized for inspection of the original review, raw and adjusted ratings, and topic-level audit trail, supporting pedagogical reflection informed by analytics, and prompting further investigation where appropriate (Tsai & Martinez-Maldonado, 2022; Gašević et al., 2015).

The framework makes its evaluative taxonomy explicit and configurable. Categories can be defined around a stated purpose and local learning design (Lockyer et al., 2013), making visible the dimensions intended to inform a particular interpretation of teaching feedback and those considered separately.

At an aggregate level, topic-level emotional profiles may support exploratory identification of recurring concerns across courses or departments, provided they are interpreted in context rather than used as instructor rankings or performance measures.

5.3 Boundaries, Risks, and Non-Use

Additional information is not inherently safer to act on. The framework produces model-based adjustments under a stated pedagogical taxonomy. An attenuated rating is not a validated estimate of teaching quality; the framework decomposes a review into topic and emotion components, it does not summarize or replace the review itself. Any interpretation that bypasses the original text discards exactly the context the audit trail is designed to preserve.

Auditability alone does not make a deployment trustworthy. Drachsler and Greller's DELICATE framework treats transparency as one element of trusted learning analytics, alongside an explicit purpose, stakeholder involvement, data governance, and routes to contest harmful consequences (Drachsler & Greller, 2016). The mechanism can reproducibly compute review-level decompositions and adjustments; it must not autonomously rank, reward, penalize, or otherwise make consequential decisions about instructors or students. The present proof of concept does not authorize deployment in a consequential institutional or platform decision process.

The framework addresses one potential source of construct-irrelevant variance: taxonomy-excluded affect. Other documented sources of bias in teaching evaluations, including gender, race, and popularity effects (Kim et al., 2025; Kopciuszewska & Rybinski, 2025), remain present in the data and in the adjusted ratings. A user who assumes that attenuation has produced a "cleaned" rating risks greater confidence in a result that still carries unaddressed contamination.

Responsible use requires understanding where the framework's auditability ends. The attenuation rule itself is fully transparent: $\zeta = 1 - D_{\text{misc}}$ is a fixed, monotonic function with no learned parameters. However, the components that produce its inputs are not equally interpretable. Topic assignment relies on sentence embeddings and cosine similarity thresholds, emotion classification uses a fine-tuned RoBERTa model, and the downstream rating model is a CatBoost ensemble. The audit trail shows *what* the framework did — which clauses were classified where, what emotions were detected, how the rating changed — but a user contesting an adjustment must recognize that the upstream classifications themselves are model outputs, not ground-truth labels. Engaging critically with the trail, rather than accepting it as a complete explanation, is part of using the framework responsibly.

6. Limitations and Future Work

This study has no independent external criterion against which to compare whether adjusted ratings support more defensible inferences about teaching than raw ratings. Mechanistic coherence, robustness checks, and expert paired- comparison agreement are indirect evidence; they do not demonstrate that adjusted ratings more accurately represent teaching quality. A stronger validity argument would require pre-specified, triangulated external evidence, for example, appropriately contextualized peer review, course-design evidence, and student-learning measures, evaluated on independent datasets. Such evidence is unavailable in the RMP corpus.

The exclusive use of anonymous, self-selected RMP reviews limits generalizability to formal SET, and the corpus's 5-star skew leaves fewer low-rating reviews for modelling. ATC relies on broad, single-label topics (strict accuracy 0.72), so multi-topic clauses may be misclassified, and the RoBERTa-GoEmotions model, trained on Reddit comments, may misinterpret sarcasm or informal language. Imperfect cleaning and clause segmentation can propagate errors downstream.

Attenuation outcomes depend on upstream classifications and the fitted model: high D_{misc} does not guarantee a large adjustment, and wholly *Miscellaneous* reviews ($D_{\text{misc}} = 1$) receive identical attenuated predictions ($\hat{y} = 3.65$) regardless of emotional content. The word-count-based D_{misc} is only a proxy for relevance; future work could incorporate semantic similarity or other textual-quality measures. Because the adjusted rating adds Δ to the raw rating without clipping, 1,643 reviews (9.6%) fall outside the original 1–5 scale, ranging from -0.31 to 7.62. These values function as unbounded model-based summaries for inspection rather than replacement scale scores.

Because the taxonomy may both omit pedagogically relevant concerns and miss other sources of construct-irrelevant variance, taxonomy-excluded affect is neither a claim that *Miscellaneous* content is intrinsically irrelevant nor a comprehensive measure of construct-irrelevant variance. Accordingly, attenuation cannot be interpreted as a comprehensive correction for construct-irrelevant variance. Demographic information in the source dataset was not used, so this analysis cannot assess whether adjusted ratings retain gender or racial inequities.

The three-expert evaluation concerns relative review judgments on a small, curated pair set, rather than exact adjustment magnitudes. Agreement remains conditional on the panel and selection protocol, including informal author review of ATC assignments during pair construction; the expert sample spans training and held-out instructors. Wilson intervals condition on these experts and assume independent pairs; the 144 distinct instructors and absence of within-pair instructor overlap support this approximation, though eight instructors recur across pairs. The unanimity breakdown is post hoc, and the 0.5 contrast cutoff defines the evaluated groups rather than an estimated discrimination threshold. Permutation interpretations remain conditional on the specified density shuffle and the model-based pair selection.

Future work should co-design and evaluate a formal-SET version of the framework with educators, students, and institutional stewards, using independent datasets and external teaching-quality criteria in line with the evaluative standards proposed by Ferguson and Clow (2017). Formal SET's structured prompts and standardized Likert scales may alter the distribution of *Miscellaneous* content, requiring recalibration. The formal-SET version should ground finer taxonomies in course learning design, potentially align them with SEEQ (Marsh, 1982), and use multi-label or soft assignment to distinguish, for example, procedural from interactional fairness and better represent multi-topic clauses. Before deployment can be considered, its purpose, access, retention, and contestation arrangements must be specified, and fairness audits should assess whether it amplifies inequities across instructor demographics and disciplines.

7. Conclusion

Open-ended teaching feedback is an increasingly common learning-analytics trace, yet computational approaches to it remain largely descriptive. This research presented validity-weighted attenuation, an auditable mechanism, motivated by validity theory and robust-estimation principles, that attenuates the modelled contribution of taxonomy-excluded affect to a rating summary, offered as a complement to existing approaches for interpreting open-ended teaching feedback.

Its contributions are an explicit attenuation mechanism with review-level audit trails; a proof-of-concept characterization on 17,127 RMP reviews, including held-out-instructor analyses and robustness checks; and an initial expert-comparison protocol with a density-only reference result: density alone matched the full mechanism's aggregate expert-ranking agreement on the sampled pairs. This comparison shows no aggregate expert-ranking advantage for attenuation on the sampled pairs. Beyond relevance ordering, the framework supports examination of the rating model's learned relationships between topic-specific emotion features and ratings, and traces the direction and magnitude of its response to attenuating taxonomy-excluded emotion features.

The framework preserves every original review and its full emotional profile; attenuation reweights the modelled contribution of affect to a summary rating. This design reflects a commitment to contestability: stakeholders can inspect and question the basis of any adjustment. As a proof of concept, the work makes validity concerns computationally explicit and auditable under a stated taxonomy, providing a mechanism, an evaluation protocol, and a density-only baseline for future work on validity-aware teaching-feedback analytics.

Declaration of Conflicting Interest

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Ethical Considerations

All expert labelling was conducted under a REB exemption approved by the authors' institution, ensuring that annotation procedures adhered to ethical research standards.

Data & Code Availability

A public codebase for this framework is available online. Due to privacy restrictions, the original RMP dataset is not included; however, all fully anonymized expert-labelled data used for validation is provided.

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